

COMS 2300 HW3

1) Proof 1: Direct Proof

Statement: If p is odd, then $p^3 + p^2 + 5p$ is also odd

Let us assume that p is an odd integer, so for some integer k , p can be written as:

$$p = 2k+1$$

So p^2 can be written as:

$$p^2 = (2k + 1)^2$$

$$p^2 = 4k^2 + 1 + 4k$$

p^3 can be written as:

$$p^3 = (2k + 1)(2k + 1)(2k + 1)$$

$$p^3 = (4k^2 + 4k + 1)(2k + 1)$$

$$p^3 = 2k(4k^2 + 4k + 1) + 4k^2 + 4k + 1$$

$$p^3 = 8k^3 + 12k^2 + 6k + 1$$

Multiply p by 5:

$$5p = 10k + 5$$

Over here, $10k$ will always be even and $5p$ will always be odd for any integers k and p .

Similarly, we can say that p^2 and p^3 are also because we can infer from the above equations because any numbers that are multiplied with 4,6,8 and 12 will always be even and their sums will be even as well, but if we add 1 to each equation in the end, then the total sum will result in an odd integer.

Hence proved, that if p is odd then, $p^3 + p^2 + 5p$ is also odd.

Proof 2: Contrapositive Proof

Statement: If $p^3 + p^2 + 5p$ is even, then p is even

Let us assume that $p^3 + p^2 + 5p$ is even. So it can be represented as:

$$p^3 + p^2 + 5p = 2k$$

Since the sum of any 3 even integers is always even, we can say that p^3 , p^2 and $5p$ are even. Since $5p$ is even, we can say that p is even because when we multiply with an even number, then only we get an even product otherwise we get an odd product. Secondly, we also know that the cube and squares of an even number are always even, so from there also we can infer that p is even.

2) Contradiction rule:

Reason: In my opinion it is difficult to prove this problem by direct proof because we need assume that if $3x$ is irrational, then $x/5$ is irrational and a little time consuming as well, however, in this case we just need to prove that the opposite of the conclusion does not comply with our assumption and so the original statement becomes true, which is easy and very efficient in my opinion.

Statement: If $3x$ is irrational, then $x/5$ is also rational.

Let us assume that $x/5$ is rational.

Any rational can be written in the following form:

$$a/b$$

So, that means:

$$x/5 = a/b$$

$$x = 5a/b$$

Multiply both sides by 3

$$3x = 15a/b$$

Since $15a$ and b are integers, so we can say that $3x$ is a rational number, but this contradicts our assumption that $3x$ is irrational and that means that the statement $x/5$ is a rational number is also false.

Hence, we can say that if $3x$ is irrational, then $x/5$ is also irrational.

3) Statement: If x, y and z are rational numbers, then $x+yz$ is also rational.

Direct Proof:

Since x, y and z are rational numbers, so they can be written in the form of:

$$x=a/b, y=p/q \text{ and } z=r/s$$

Where a, b, p, q and r, s are integers and b, q and s are non-zero integers

So,

$$\begin{aligned} x+yz &= a/b + (p/q * r/s) \\ &= a/b + pr/qs \\ &= (aqs+bpr) / bqs \end{aligned}$$

Since a, q, s and b, p, r are all integers and b, q, s are not equal to 0, we can say that $x + yz$ is a rational number since it can be represented in the form of a/b .

Hence proved, if x, y and z are rational numbers then $x+yz$ is also a rational number.

4) Contradiction rule

Reason: We can't use a direct proof over here since we are not allowed to use the theorems for rational and irrational numbers, so that's I have used a contradiction rule over here

Statement: If x, y are non zero rational numbers and z is an irrational number, then $x + yz$ must be a rational number.

Since, $x+yz$ is a rational number we can say that:

$$x+yz=r$$

Where r is a rational number.

So, we can write z as:

$$z=(r-x)/y$$

Since r, x and y are rational numbers we can say that z is a rational number as well, however, this contradicts our assumption that z is irrational, so that means our statement is false.

Thus, we can say that $x+yz$ is irrational.

5) Contradiction rule

Statement: If $mn \geq 49$, then $m < 7$ and $n < 9$

Since m and n are less than 7 and 9 respectively, so we can say that:

$$\begin{aligned} m &= 1, 2, 3, 4, 5, 6 \\ n &= 1, 2, 3, 4, 5, 6, 7, 8 \end{aligned}$$

So, the largest possible product of m and n is:

$$mn = 6 \cdot 8 = 48$$

Since, $48 < 49$, we can say that our statement is false because this conclusion contradicts our assumption that $mn \geq 49$.

Hence, we can say that if $mn \geq 49$, then $m \geq 7$ and $n \geq 9$.

6) Let:

F = number of freshmen
 S = number of sophomores
 J = number of juniors
 X = number of seniors

By Contradiction rule:

Statement: If the college organization has 36 students, then $F < 9$, $S < 8$, $J < 12$ and $X < 10$.

So the maximum number of students that the organization can have is:

$$8+7+11+9 = 35$$

Since $35 < 36$, then we can say that our statement is false, since we assumed that the number of students in the organization are 36.

Hence, we can say that If the college organization has 36 students, then $F \geq 9$, $S \geq 8$, $J \geq 12$ and $X \geq 10$.

7) Statement: If $p \geq 6$ or $p \leq -12$, then $(p + 3)^2 \geq 81$

We have 2 different cases over here:

- 1) If $p \geq 6$, then $(p + 3)^2 \geq 81$
- 2) If $p \leq -12$, then $(p + 3)^2 \geq 81$

Case 1: $p \geq 6$

Add 3 to both the sides:

$$\begin{aligned} p + 3 &\geq 6 + 3 \\ p + 3 &\geq 9 \end{aligned}$$

Squaring both the sides:

$$(p + 3)^2 \geq 81$$

Hence proved, if $p \geq 6$ then $(p + 3)^2 \geq 81$.

Case 2: $p \leq -12$

Add 3 to both the sides:

$$\begin{aligned} p + 3 &\leq -12 + 3 \\ p + 3 &\leq -9 \end{aligned}$$

Squaring both the sides:

$$(p + 3)^2 \geq 81$$

Note: Since we are squaring the sides for a negative the inequality sign reverses itself

Hence proved, if $p \leq -12$ then $(p + 3)^2 \leq 81$.

Since we were able to prove both the cases, we can say that if $p \geq 6$ or $p \leq -12$, then $(p + 3)^2 \geq 81$.

8) **Statement:** If x is an irrational number and y is a rational number then, x^y is a rational number.

If we say that $x = \sqrt{2}$ and $y = 0$, then:

$$x^y = \sqrt{2}^0 = 1$$

We got 1 as the answer because when any number, whether rational or irrational, is raised to the 0 then we get the answer as 1 by the rule of exponents. Also since we have defined our x and y to be $\sqrt{2}$ and 0 and then we went on to prove the statement. We are also not relying on any kind of logical reasoning (such as contrapositive, contradiction or direct proofs) to show that such numbers might exist without providing any example, so that's why we can say that my proof is constructive.